

NEW LAGUERRE HISTORY INTERPRETATIONS OF EULER NUMBERS AND REDUCED TANGENT NUMBERS

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ABSTRACT. In this paper, we introduce two families of partial Laguerre histories by imposing natural restrictions on the labels of matched north-east and south-east steps. We show that these two families share the same Jacobi continued-fraction expansion. As applications, we derive new combinatorial interpretations of the Euler numbers and the reduced tangent numbers by excluding certain colored east steps. We also introduce a natural (p, q) -weight on partial Laguerre histories and obtain (p, q) -analogues of Euler numbers and reduced tangent numbers.

1. INTRODUCTION

The pioneer work on analytic continued fractions is due to Leonhard Euler, who introduced several expansions of power series, such as the one in [Eul60]:

$$\sum_{n \geq 0} (n+1)! z^n = \frac{1}{1 - 2z - \frac{1 \cdot 2 z^2}{1 - 4z - \frac{2 \cdot 3 z^2}{\ddots - \frac{k(k+1) z^2}{1 - 2kz - \ddots}}}}. \quad (1.1)$$

The general theory of such continued fractions was primarily developed by Stieltjes in his seminal memoir [Sti94]. These continued fractions are a particular case of the so-called *Jacobi continued fractions*, or *J-fractions*, that is,

$$\mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) := \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{\ddots - \frac{\lambda_{k+1} z^2}{\ddots}}}}, \quad (1.2)$$

where $\mathbf{b} := (b_k)_{k \geq 0}$ and $\boldsymbol{\lambda} := (\lambda_k)_{k \geq 1}$ denote two sequences of “coefficients,” which may consist of integers, real or complex numbers, or, more generally, of formal variables drawn from two distinct countably infinite alphabets. The *Euler number* E_n is the n -th coefficient in the Taylor expansion of $\sec(z) + \tan(z)$, namely,

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z). \quad (1.3)$$

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The Euler numbers have been widely studied in Combinatorics [Vie81; Han20; LYZ25; Sok20; Jos14; FLS25]. It is well-known that the J -fraction for Euler numbers E_{n+1} :

$$\sum_{n=0}^{\infty} E_{n+1} z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (1.4)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by

$$b_k = k + 1 \text{ and } \lambda_{k+1} = \binom{k+2}{2} \text{ for } k \geq 0. \quad (1.5)$$

The continued fraction (1.4) is classical and follows as a specialization of a continued fraction formula due to Stieltjes [Sti94], as pointed out by Zeng in [Sok20, Note 9]. It was rediscovered empirically by P. D. Hanna in a contribution on [Slo25, A000111] and later proved by a combinatorial method of Josuat-Vergès [Jos14]. Han [Han20] considered a q -version of (1.4) (See Eq. (3.20) with $p = 1$) and asked for a combinatorial interpretation of the corresponding q -Euler numbers, which was later provided by Pan and Zeng [PZ21] using André permutations. Our Theorem 2.3 gives more combinatorial interpretations in terms of partial Laguerre histories.

The odd-indexed Euler numbers E_{2n+1} are called the *tangent numbers*, and the *reduced tangent numbers* are the normalized tangent numbers defined by

$$R_n := \frac{E_{2n+1}}{2^n} \quad (n \geq 0). \quad (1.6)$$

These numbers R_n admit the following J -fraction:

$$\sum_{n \geq 0} R_n z^{2n} = \mathcal{J}(z; \mathbf{0}, \boldsymbol{\lambda}), \quad (1.7)$$

where $\boldsymbol{\lambda}$ is given by $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. The above J -fraction follows from a classical formula of Stieltjes, as pointed out by Bala [Bal17]. The aim of this paper is to give Laguerre-history interpretations of the Euler numbers and the reduced tangent numbers, and to extend them to natural q -analogues. In Section 2, we introduce two restricted families of Laguerre histories, called r -partial and l -partial Laguerre histories, and show that forbidding suitable colored east steps recovers the two continued fractions above. In Section 3, we define two statistics on partial Laguerre histories and derive combinatorial interpretations of the resulting (p, q) -Euler numbers and (p, q) -reduced tangent numbers.

2. PARTIAL LAGUERRE HISTORY

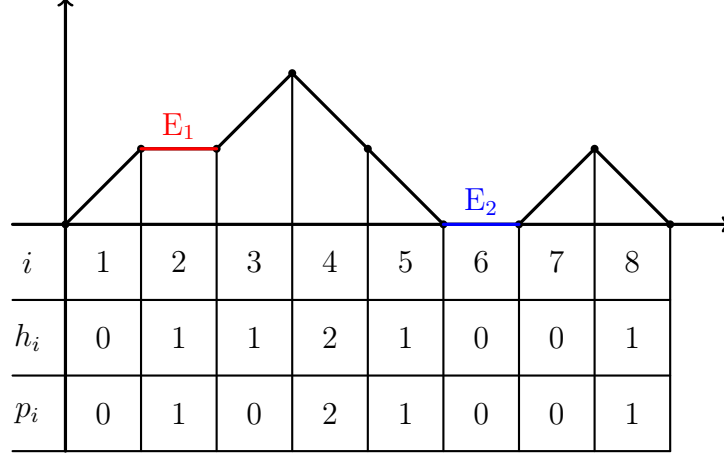
A *Motzkin path* of length n is a lattice path $\{(x_i, y_i)\}_{0 \leq i \leq n}$ in the plane that

- starts at $(x_0, y_0) = (0, 0)$ and ends at $(x_n, y_n) = (n, 0)$;
- uses only the three types of steps

$$N = (1, 1) \text{ (North-East)}, \quad S = (1, -1) \text{ (South-East)}, \quad E = (1, 0) \text{ (East)};$$

- never goes below the x -axis, that is, the height $h_i := y_{i-1} \geq 0$ for $i = 1, \dots, n$.

Let $\mathcal{M}_n$ denote the set of Motzkin paths of length n . Following the classical Flajolet's continued fraction approach, we consider a *weighted Motzkin-path model*. Each East step at height i is assigned a weight b_i , while each North-East (resp. South-East) step at height i receives a weight a_i (resp. c_i). The weight of a path $P \in \mathcal{M}_n$ is the product of the weights of its steps and is denoted by $\omega(P)$. The generating function of weighted Motzkin

FIGURE 1. A Laguerre history in $\mathcal{L}_8$.

paths admits a J -fraction expansion, given by Flajolet's fundamental lemma on continued fractions [Fla80].

Lemma 2.1. *We have*

$$\sum_{n \geq 0} \sum_{P \in \mathcal{M}_n} \omega(P) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.1)$$

where $\mathbf{b} = (b_k)_{k \geq 0}$ and $\boldsymbol{\lambda} = (a_{k-1}c_k)_{k \geq 1}$.

We can identify a Motzkin path of length n by a *Motzkin word* $w = w_1 \dots w_n$ such that $w_i = \text{N}$ (resp. S, E) if and only if the i th step is N (resp. S, E). Given a Motzkin path $w = w_1 w_2 \dots w_n \in \mathcal{M}_n$, the height h_i of the i th step is given by

$$h_i := h_i(w) = \#\{j < i : w_j = \text{N}\} - \#\{j < i : w_j = \text{S}\}. \quad (2.2)$$

A *2-Motzkin path* is a Motzkin path whose east steps could come in two colors: red ($\mathbf{E}_1$) or blue ($\mathbf{E}_2$). A *Laguerre history* of length n , introduced by Viennot [Vie83], is a couple (w, p) , where $w = w_1 \dots w_n$ is a 2-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is a label sequence such that $0 \leq p_i \leq h_i$ for $1 \leq i \leq n$. Denote by $\mathcal{L}_n$ the set of Laguerre histories of length n , see Fig. 1 for a Laguerre history in $\mathcal{L}_8$. Françon and Viennot [FV79] proved the cardinality of $\mathcal{L}_n$ is $(n+1)!$ by constructing a bijection from symmetric group $\mathfrak{S}_{n+1}$ to $\mathcal{L}_n$.

Let $w = w_1 \dots w_n \in \mathcal{M}_n$ be a Motzkin path. For each South-East step $w_j = \text{S}$, its matching North-East step is the unique step $w_i = \text{N}$, with $i < j$, such that

$$h_j = h_i + 1, \quad (2.3)$$

and every step w_k , for $i < k < j$ satisfies $h_k > h_i$. The pair (i, j) is called a *matched pair*. We denote by $\text{MP}(w)$ the set of all such matched pairs of w . A Laguerre history $(w, p) \in \mathcal{L}_n$ is said to be a *r-partial* (resp. *l-partial*) *Laguerre history* if each matched pair $(i, j) \in \text{MP}(w)$, with $i < j$ satisfies

$$p_i \geq p_j \quad (\text{resp. } p_i < p_j). \quad (2.4)$$

Let $\mathcal{L}_n^r$ (resp. $\mathcal{L}_n^l$) be the set of all r -partial (resp. l -partial) Laguerre histories of length n . Throughout this paper, $|S|$ denotes the cardinality of a finite set S . For a Laguerre history

$(w, p) \in \mathcal{L}_n$, we define

$$e_1(w, p) := |\{i \in [n] : w_i = \mathbf{E}_1\}| \text{ and } e_2(w, p) := |\{i \in [n] : w_i = \mathbf{E}_2\}|, \quad (2.5)$$

and the following two enumerative polynomials

$$L_n^l(x, y) := \sum_{\omega := (w, p) \in \mathcal{L}_n^l} x^{e_1(\omega)} y^{e_2(\omega)} \text{ and } L_n^r(x, y) := \sum_{\omega := (w, p) \in \mathcal{L}_n^r} x^{e_1(\omega)} y^{e_2(\omega)}. \quad (2.6)$$

The following theorem is our main result.

Theorem 2.2. *For $s \in \{l, r\}$, we have*

$$\sum_{n \geq 0} L_n^s(x, y) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.7)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = (x + y)(k + 1)$, and $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$, respectively.

Proof. We prove the statement for $s = r$; the case $s = l$ is similar. Fix an underlying Motzkin path $w \in \mathcal{M}_n$. We count all ways to turn w into an r -partial Laguerre history (w, p) , keeping track of the two colors of east steps by the variables x and y .

First consider an east step of w at height k . In a Laguerre history, this step can be colored either $\mathbf{E}_1$ or $\mathbf{E}_2$, and its label can be chosen freely among $0, 1, \dots, k$. Therefore such a step contributes

$$(k + 1)x + (k + 1)y = (x + y)(k + 1). \quad (2.8)$$

Next consider a matched pair $(i, j) \in \text{MP}(w)$, where $w_i = \text{N}$, $w_j = \text{S}$, and the step N starts at height k . Then the label of the step N satisfies $0 \leq p_i \leq k$, while the label of the step S satisfies $0 \leq p_j \leq k + 1$. Since (w, p) is required to be r -partial, we impose $p_i \geq p_j$. Thus the number of admissible choices for the two labels is

$$\sum_{p_i=0}^k (p_i + 1) = \binom{k+2}{2}. \quad (2.9)$$

Hence the generating polynomial for r -partial Laguerre histories is the generating function of weighted Motzkin paths with weights

$$b_k = (x + y)(k + 1), \quad a_k = 1, \quad c_{k+1} = \binom{k+2}{2} \quad (k \geq 0). \quad (2.10)$$

Indeed, an east step at height k contributes b_k , while a matched pair consisting of a north-east step starting at height k and its matched south-east step contributes

$$\lambda_{k+1} := a_k c_{k+1} = \binom{k+2}{2}, \quad (2.11)$$

which is exactly the number of admissible label choices computed in (2.10). By Lemma 2.1, this gives (2.7) for $s = r$.

For $s = l$, the number of admissible labels for matched pairs is

$$\#\{(p_i, p_j) : 0 \leq p_i \leq k, 0 \leq p_j \leq k + 1, p_i < p_j\}, \quad (2.12)$$

which is equal to

$$\sum_{p_i=0}^k (k + 1 - p_i) = \binom{k+2}{2}. \quad (2.13)$$

Thus, for l -partial Laguerre histories, each matched north-east/south-east pair starting at height k has the same weight $\binom{k+2}{2}$ as in the r -partial case, while the contribution of east steps is unchanged. Applying Lemma 2.1 again, we obtain (2.7) for $s = l$ as well. This completes the proof. $\square$

For $s \in \{r, l\}$ and $I \subseteq \{1, 2\}$, we define

$$\mathcal{L}_n^{s,I} := \{(w, p) \in \mathcal{L}_n^s : e_i((w, p)) = 0 \text{ for all } i \in I\}. \quad (2.14)$$

Theorem 2.3. *For $s \in \{r, l\}$, and $\epsilon \in \{1, 2\}$, we have*

$$\sum_{n \geq 0} |\mathcal{L}_n^{s,\{\epsilon\}}| z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.15)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = k + 1$ and $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. Consequently, we have

$$|\mathcal{L}_n^{s,\{\epsilon\}}| = E_{n+1}. \quad (2.16)$$

Proof. By Theorem 2.2, the specializations $(x, y) = (0, 1)$ and $(x, y) = (1, 0)$ give, respectively, the generating functions of $\mathcal{L}_n^{s,\{1\}}$ and $\mathcal{L}_n^{s,\{2\}}$. In both cases, the resulting coefficients are $b_k = k + 1$ and $\lambda_{k+1} = \binom{k+2}{2}$. By the continued fraction (1.4), both generating functions are equal to $\sum_{n \geq 0} E_{n+1} z^n$. It follows from comparing their coefficients of z^n that (2.16). $\square$

Theorem 2.4. *For $s \in \{r, l\}$, we have*

$$\sum_{n \geq 0} |\mathcal{L}_{2n}^{s,\{1,2\}}| z^{2n} = \mathcal{J}(z; \mathbf{0}, \boldsymbol{\lambda}), \quad (2.17)$$

where $\boldsymbol{\lambda}$ is given by $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. Consequently, we have

$$|\mathcal{L}_{2n}^{s,\{1,2\}}| = R_n. \quad (2.18)$$

Proof. By Theorem 2.2, setting $x = y = 0$ gives the generating function of r -partial and l -partial Laguerre histories with no east steps, that is, $\mathcal{L}_n^{l,\{1,2\}}$ and $\mathcal{L}_n^{r,\{1,2\}}$. Since such paths contain only north-east and south-east steps, their length must be even. Then, comparing this with the continued fraction (1.7) for the reduced tangent numbers gives (2.18). $\square$

3. A (p, q) -ANALOGUE

In this section, we give a natural (p, q) -analogue of Theorem 2.2. Let $(w, p) \in L_n^s$, where $s \in \{r, l\}$. Recall that $\text{MP}(w)$ denotes the set of matched north-east/south-east pairs of the underlying Motzkin path w . If $(i, j) \in \text{MP}(w)$, then $w_i = \text{N}$, $w_j = \text{S}$. For $s \in \{l, r\}$, we define two statistics area_s and coarea_s on two kinds of partial Laguerre histories as follows, respectively. For $(w, p) \in \mathcal{L}_n^r$, define

$$\text{area}_r(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} p_i + \sum_{(i,j) \in \text{MP}(w)} (p_i + p_j), \quad (3.1)$$

and

$$\text{coarea}_r(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} (h_i - p_i) + \sum_{(i,j) \in \text{MP}(w)} (2h_i - p_i - p_j). \quad (3.2)$$

For $(w, p) \in \mathcal{L}_n^l$, define

$$\text{area}_l(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \textcolor{red}{E}_1 \text{ or } \textcolor{blue}{E}_2}} p_i + \sum_{(i,j) \in \text{MP}(w)} (p_i + p_j - 1), \quad (3.3)$$

and

$$\text{coarea}_l(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \textcolor{red}{E}_1 \text{ or } \textcolor{blue}{E}_2}} (h_i - p_i) + \sum_{(i,j) \in \text{MP}(w)} (2h_i + 1 - p_i - p_j). \quad (3.4)$$

For $s \in \{r, l\}$, define the following enumerative polynomials,

$$L_n^s(x, y; p, q) := \sum_{\omega = (w, p) \in \mathcal{L}_n^s} x^{e_1(\omega)} y^{e_2(\omega)} p^{\text{coarea}_s(\omega)} q^{\text{area}_s(\omega)}. \quad (3.5)$$

The (p, q) -analogue of n is defined by

$$[n]_{p,q} = \frac{p^n - q^n}{p - q} = \sum_{i+j=n-1} p^i q^j \quad (n \in \mathbb{N}),$$

and the (p, q) -analogue of the binomial coefficient $\binom{n}{k}$ is defined by

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{p,q} = \frac{[n]_{p,q} [n-1]_{p,q} \cdots [n-k+1]_{p,q}}{[1]_{p,q} [2]_{p,q} \cdots [k]_{p,q}} \quad (0 \leq k \leq n).$$

Recall [Sta99, Chapter 7] the standard specialization of the complete homogeneous symmetric function of degree k

$$h_k(1, q, \dots, q^n) = \left[\begin{matrix} n+k \\ k \end{matrix} \right]_q. \quad (3.6)$$

Lemma 3.1. *For $n, k \geq 0$, we have*

$$h_k(p^n, p^{n-1}q, \dots, pq^{n-1}, q^n) = \left[\begin{matrix} n+k \\ k \end{matrix} \right]_{p,q}. \quad (3.7)$$

Proof. By homogeneity, replacing q by q/p , we obtain

$$h_k(p^n, p^{n-1}q, \dots, pq^{n-1}, q^n) = p^{nk} h_k\left(1, \frac{q}{p}, \dots, \left(\frac{q}{p}\right)^n\right). \quad (3.8)$$

By (3.6), we derive

$$h_k(p^n, p^{n-1}q, \dots, pq^{n-1}, q^n) = p^{nk} \left[\begin{matrix} n+k \\ k \end{matrix} \right]_{q/p}. \quad (3.9)$$

Since $[m]_{p,q} = p^{m-1} [m]_{q/p}$, it follows directly from the definition of Gaussian binomial coefficients that

$$\left[\begin{matrix} n+k \\ k \end{matrix} \right]_{p,q} = p^{nk} \left[\begin{matrix} n+k \\ k \end{matrix} \right]_{q/p}. \quad (3.10)$$

This proves (3.7). $\square$

Theorem 3.2. For $s \in \{r, l\}$, we have

$$\sum_{n \geq 0} L_n^s(x, y; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (3.11)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by

$$b_k = (x + y)[k + 1]_{p,q} \text{ and } \lambda_{k+1} = \begin{bmatrix} k + 2 \\ 2 \end{bmatrix}_{p,q} \quad (k \geq 0). \quad (3.12)$$

Proof. We first prove the statement for $s = r$. Fix an underlying Motzkin path $w \in \mathcal{M}_n$. We count all ways to turn w into an r -partial Laguerre history (w, p) , keeping track of the colors of east steps by x, y , and of the labels by the statistics area_r and coarea_r .

Consider an east step of w at height k . Its label can be chosen freely among $0, 1, \dots, k$, and it can be colored either $\mathbf{E}_1$ or $\mathbf{E}_2$. If the label is p_i , then this east step contributes $p^{k-p_i} q^{p_i}$. Hence the total contribution of such an east step is

$$x \sum_{a=0}^k p^{k-a} q^a + y \sum_{a=0}^k p^{k-a} q^a = (x + y)[k + 1]_{p,q}. \quad (3.13)$$

Thus, we have $b_k = (x + y)[k + 1]_{p,q}$.

Next consider a matched pair $(i, j) \in \text{MP}(w)$, where $w_i = \text{N}$, $w_j = \text{S}$, and the north-east step starts at height k . In the r -partial case, the labels satisfy

$$0 \leq p_j \leq p_i \leq k. \quad (3.14)$$

Moreover, this matched pair contributes $p^{2k-p_i-p_j} q^{p_i+p_j}$ to $p^{\text{coarea}_r} q^{\text{area}_r}$. Hence the total contribution of the admissible label choices is

$$\sum_{p_i=0}^k \sum_{p_j=0}^{p_i} p^{2k-p_i-p_j} q^{p_i+p_j} = \begin{bmatrix} k + 2 \\ 2 \end{bmatrix}_{p,q}. \quad (3.15)$$

Indeed, by setting $r = p_j$, $s = p_i - p_j$, and $t = k - p_i$, the pairs (p_i, p_j) satisfying $0 \leq p_j \leq p_i \leq k$ are in bijection with the triples (r, s, t) satisfying $r, s, t \geq 0$ and $r + s + t = k$. Moreover,

$$p^{2k-p_i-p_j} q^{p_i+p_j} = (q^2)^r (pq)^s (p^2)^t.$$

Therefore, we have

$$\sum_{p_i=0}^k \sum_{p_j=0}^{p_i} p^{2k-p_i-p_j} q^{p_i+p_j} = h_k(p^2, pq, q^2).$$

Combining this and Lemma 3.1 with $n = 2$, and $\begin{bmatrix} k+2 \\ k \end{bmatrix}_{p,q} = \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_{p,q}$, we obtain (3.15). In the notation of Lemma 2.1, we choose

$$b_k = (x + y)[k + 1]_{p,q}, \quad a_k = 1, \quad c_{k+1} = \begin{bmatrix} k + 2 \\ 2 \end{bmatrix}_{p,q} \quad \text{for } k \geq 0. \quad (3.16)$$

Applying Lemma 2.1 gives the desired continued fraction for $s = r$.

The case $s = l$ is similar. In this case, the labels of a matched pair (i, j) starting at height k satisfy

$$0 \leq p_i < p_j \leq k + 1. \quad (3.17)$$

The contribution of this matched pair to $p^{\text{coarea}_l} q^{\text{area}_l}$ is $p^{2k+1-p_i-p_j} q^{p_i+p_j-1}$. Therefore its total contribution is

$$\sum_{p_i=0}^k \sum_{p_j=p_i+1}^{k+1} p^{2k+1-p_i-p_j} q^{p_i+p_j-1} = \left[\begin{matrix} k+2 \\ 2 \end{matrix} \right]_{p,q}. \quad (3.18)$$

By the change of variables $u = p_i$ and $v = p_j - 1$, the sum becomes

$$\sum_{v=0}^k \sum_{u=0}^v p^{2k-u-v} q^{u+v}, \quad (3.19)$$

which is the same sum as in Eq. (3.15). Thus, it follows (3.18). Applying Lemma 2.1 again, with the same weight (3.16), gives the desired continued fraction for $s = l$. $\square$

We define (p, q) -analogues $E_n(p, q)$ and $R_n(p, q)$ of the Euler numbers and the reduced tangent numbers by

$$\sum_{n \geq 0} E_{n+1}(p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) \quad (3.20)$$

and

$$\sum_{n \geq 0} R_n(p, q) z^{2n} = \mathcal{J}(z; 0, \boldsymbol{\lambda}) \quad (3.21)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = [k+1]_{p,q}$ and $\lambda_{k+1} = \left[\begin{matrix} k+2 \\ 2 \end{matrix} \right]_{p,q}$ for $k \geq 0$.

We now record two specializations of Theorem 3.2. They give combinatorial interpretations of $E_{n+1}(p, q)$ and $R_n(p, q)$ in terms of the statistics area_s and coarea_s on partial Laguerre histories. For $s \in \{r, l\}$, $I \subseteq \{1, 2\}$, and $n \geq 0$, define

$$L_n^{s,I}(p, q) := \sum_{\omega=(w,p) \in \mathcal{L}_n^{s,I}} p^{\text{coarea}_s(\omega)} q^{\text{area}_s(\omega)}. \quad (3.22)$$

The following two results are obtained from Theorem 3.2 by forbidding one of the two colors of east steps, or both. The first follows by comparing (3.20) with the specializations $(x, y) = (0, 1)$ and $(1, 0)$, while the second follows by comparing (3.21) with the specialization $(x, y) = (0, 0)$.

Theorem 3.3. *For $n \geq 0$, $s \in \{r, l\}$, and $\epsilon \in \{1, 2\}$, we have*

$$L_n^{s,\{\epsilon\}}(p, q) = E_{n+1}(p, q). \quad (3.23)$$

Theorem 3.4. *For $n \geq 0$ and $s \in \{r, l\}$, we have*

$$L_{2n}^{s,\{1,2\}}(p, q) = R_n(p, q). \quad (3.24)$$

Remark 3.1. *Setting $p = q = 1$ in (3.23) and (3.24) recovers (2.16) and (2.18), respectively. Setting $p = 1$ in Theorem 3.3 gives four Laguerre-history interpretations of the q -Euler numbers $E_{n+1}(q)$ considered by Han [Han20].*

4. CONCLUDING REMARKS

We finally describe the permutation classes corresponding to our partial Laguerre histories under the Françon-Viennot bijection [FV79].

Let $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, with $\sigma_0 = \sigma_{n+1} = 0$, a letter σ_i with $i \in [n]$ is called a

- *valley* if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- *peak* if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- *double descent* if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$.

For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$ and $k = \sigma_j$ with $j \in [n]$, the coordinate generalized pattern $(31-2)_k$ is defined by

$$(31-2)_k(\sigma) = |\{i : i+1 < j \text{ and } \sigma_{i+1} < k < \sigma_i\}|. \quad (4.1)$$

We describe Françon-Viennot's bijection $\Psi_{FV} : \mathfrak{S}_{n+1} \rightarrow \mathcal{L}_n$ and its inverse, see [FV79; Vie83]. For a permutation $\sigma \in \mathfrak{S}_{n+1}$, the Laguerre history $\Psi_{FV}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n$ is defined by

$$w_i = \begin{cases} \text{N (resp. S)} & \text{if } i \text{ is a valley (resp. peak) of } \sigma; \\ \text{E}_1 \text{ (resp. E}_2\text{)} & \text{if } i \text{ is a double descent (resp. double ascent) of } \sigma, \end{cases} \quad (4.2)$$

and $p_i = (31-2)_i(\sigma)$ for each $i \in [n]$.

Inversely, starting from a Laguerre history $(w, p) \in \mathcal{L}_n$ one can recover the corresponding permutation σ by the following algorithm:

- Initialization: $\sigma = \diamond$;
- At the i th ($1 \leq i \leq n$) step of the algorithm, replace the $(p_i + 1)$ th $\diamond$ (from left to right) of σ by

$$\begin{cases} \diamond i \diamond & \text{if } w_i = \text{N}; \\ i \diamond & \text{if } w_i = \text{E}_2; \\ i & \text{if } w_i = \text{S}; \\ \diamond i & \text{if } w_i = \text{E}_1. \end{cases}$$

- The final permutation is obtained by putting $n+1$ at the last remaining $\diamond$.

Example 1. Let $\sigma = 697835142 \in \mathfrak{S}_9$. We have

$$\Psi_{FV}(\sigma) = (\text{N}\text{E}_1\text{NSS}\text{E}_2\text{NS}, (0, 1, 0, 2, 1, 0, 0, 1)) \in \mathcal{L}_8,$$

see Fig. 1. Inversely, we have

$$\begin{aligned} \sigma &= \diamond \rightarrow \diamond 1 \diamond \rightarrow \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 142 \rightarrow \diamond 35142 \rightarrow 6 \diamond 35142 \\ &\rightarrow 6 \diamond 7 \diamond 35142 \rightarrow 6 \diamond 7835142 \rightarrow 697835142. \end{aligned}$$

We now translate our restrictions on Laguerre histories into conditions on permutations. Recall that, for a Motzkin path w , $\text{MP}(w)$ denotes the set of pairs (i, j) such that $i < j$, $w_i = \text{N}$, $w_j = \text{S}$, $h_j = h_i + 1$, and $h_k > h_i$ for every $i < k < j$.

Let $\sigma \in \mathfrak{S}_{n+1}$, and write $\Psi_{FV}(\sigma) = (w, p) \in \mathcal{L}_n$. We define

$$\text{MP}_{FV}(\sigma) := \text{MP}(w).$$

Thus, if $(i, j) \in \text{MP}_{FV}(\sigma)$, then i is a valley of σ , j is a peak of σ , and the corresponding N-step and S-step are matched in the Motzkin path associated with σ .

For $s \in \{r, l\}$, define $\mathfrak{S}_{n+1}^s$ to be the set of permutations $\sigma \in \mathfrak{S}_{n+1}$ such that, for every $(i, j) \in \text{MP}_{FV}(\sigma)$, one has

$$(31-2)_i(\sigma) \geq (31-2)_j(\sigma) \quad \text{if } s = r, \quad (4.3)$$

and

$$(31-2)_i(\sigma) < (31-2)_j(\sigma) \quad \text{if } s = l. \quad (4.4)$$

Then, by construction of the Françon–Viennot bijection, we have

$$\Psi_{FV}(\mathfrak{S}_{n+1}^s) = \mathcal{L}_n^s. \quad (4.5)$$

Moreover, the colors of the east steps have a simple interpretation in terms of local types of letters. The absence of $\mathbf{E}_1$ -steps is equivalent to the absence of double descents in σ , while the absence of $\mathbf{E}_2$ -steps is equivalent to the absence of double ascents in σ . Hence, Eq. (2.16) in Theorem 2.3 can be restated as

$$|\{\sigma \in \mathfrak{S}_{n+1}^s : \sigma \text{ has no double descents}\}| = E_{n+1}, \quad (4.6)$$

and

$$|\{\sigma \in \mathfrak{S}_{n+1}^s : \sigma \text{ has no double ascents}\}| = E_{n+1}. \quad (4.7)$$

Similarly, forbidding both colors of east steps means that every letter $1, \dots, 2n$ is either a valley or a peak. Therefore, Eq. (2.18) in Theorem 2.4 gives

$$|\{\sigma \in \mathfrak{S}_{2n+1}^s : \text{every letter } 1, \dots, 2n \text{ is a valley or a peak}\}| = R_n. \quad (4.8)$$

Problem 1. *Can one find an intrinsic description of the matching $\text{MP}_{FV}(\sigma)$ directly in terms of the permutation σ ? Such a description would turn Theorems 2.3 and 2.4 into purely permutation-theoretic interpretations of Euler numbers and reduced tangent numbers without referring to Laguerre histories.*

For a permutation $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, we call an index $i \in [n-1]$ a *descent* of σ if $\sigma_i > \sigma_{i+1}$. Denote by $\text{des}(\sigma)$ the number of descents of σ . Recall that the generalized pattern $31-2$ occurs in σ if there exist indices $i < j$ such that the entries in positions $i, i+1$, and j satisfy

$$\sigma_i > \sigma_j > \sigma_{i+1}.$$

Similarly, the generalized pattern $2-13$ occurs in σ if there exist indices $i < j$ such that the entries in positions i, j , and $j+1$ satisfy

$$\sigma_j < \sigma_i < \sigma_{j+1}.$$

We denote by $(31-2)(\sigma)$ and $(2-13)(\sigma)$ the numbers of occurrences of these two patterns in σ , respectively.

Pan and Zeng [PZ21] considered the following continued fraction:

$$\sum_{n \geq 0} E_{n+1}(t, p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (4.9)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = [k+1]_{p,q}$ and $\lambda_{k+1} = t \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_{p,q}$ for $k \geq 0$. They proved, by defining a group action on permutations, that polynomials $E_{n+1}(t, p, q)$ are the normalized γ -coefficients of Brändén's (p, q) -Eulerian polynomials [Brä08] in terms of André permutations. More precisely,

$$E_n(t, p, q) = \sum_{\sigma \in \mathcal{D}_n} t^{\text{des}(\sigma)} p^{(2-13)(\sigma)} q^{(31-2)(\sigma) - \text{des}(\sigma)}, \quad (4.10)$$

where $\mathcal{D}_n$ denotes the set of André permutations of the second kind of length n .

The continued fraction defined in (4.9) also arises naturally from our partial Laguerre histories. Indeed, the coefficient $\lambda_{k+1} = t \left[\begin{smallmatrix} k+2 \\ 2 \end{smallmatrix} \right]_{p,q}$ is obtained as the total weight of admissible labels on a matched pair (i, j) with $h_i = k$, where the additional factor t marks this matched pair. Thus, our model gives a natural weighted-path interpretation of the polynomials $E_{n+1}(t, p, q)$. This suggests that the partial Laguerre history model and the André permutation should be closely related. We leave the following problem for future work.

Problem 2. *Find a direct bijection between partial Laguerre histories and André permutations of the second kind. More generally, determine the precise relationship between these two combinatorial interpretations of $E_n(t, p, q)$.*

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